

30. Layer 3 Ethernet Switch Module (ESWM)

30.1 Overview

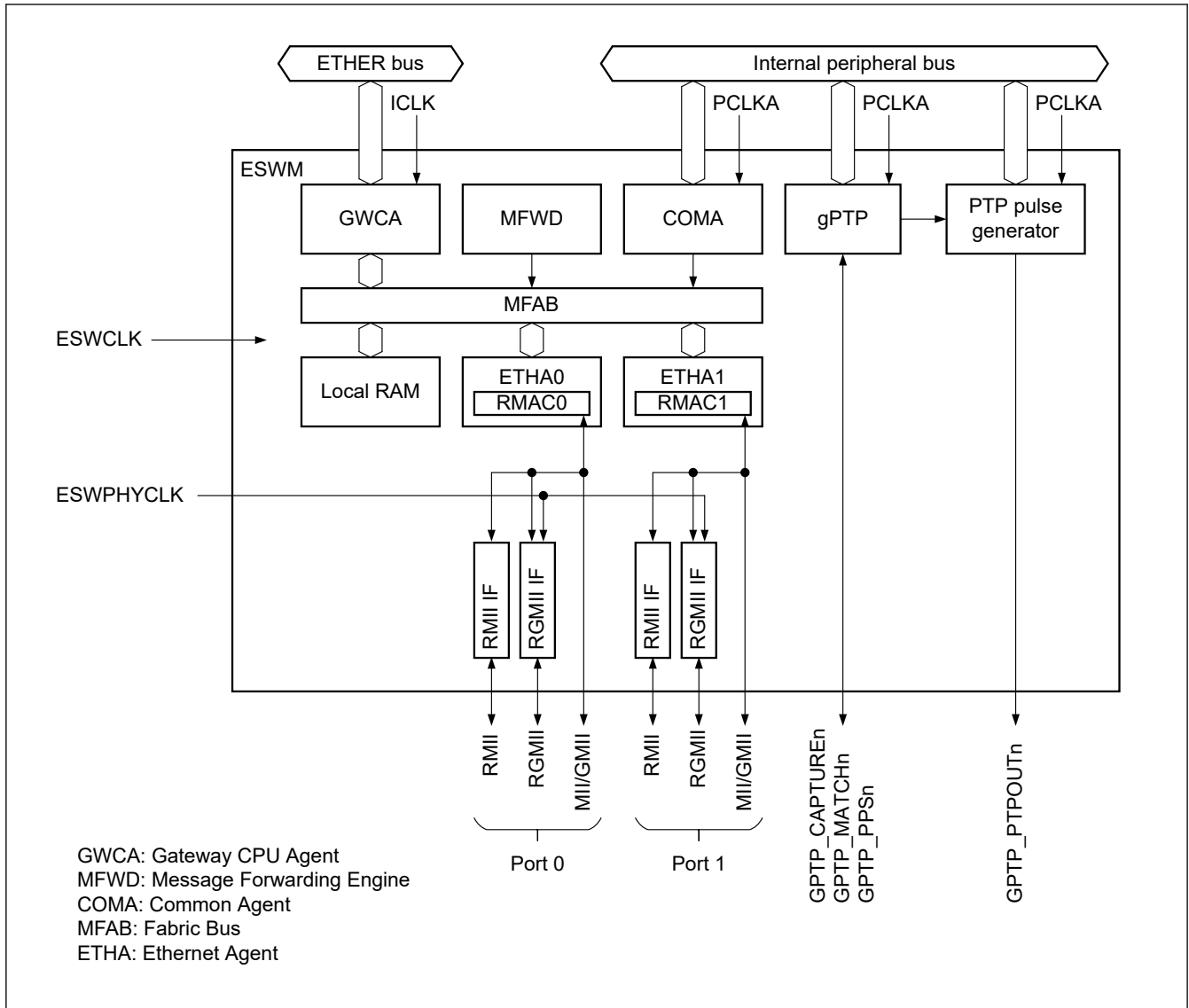
The Layer 3 Ethernet Switch Module (ESWM) consists of an Ethernet switch with higher-level routing capability and multi-protocol interface support. It allows autonomous frame routing within the same and between different network interface protocols (for now only Ethernet) for optimized gateway applications.

[Table 30.1](#) lists the ESWM specifications and [Figure 30.1](#) shows the ESWM block diagram.

Table 30.1 ESWM specifications

Item	Function		Specification
Basic function	PHY interface	Number of port	2 ports
		Type	MII, RMII, GMII, RGMII
		Speed	10 Mbps, 100 Mbps, 1 Gbps
		Mode	Full Duplex
		Management IF(MDIO)	Yes
		Energy Efficient Ethernet (low power idle)	Yes
		Magic Packet™	Yes
	Switch function	Layer 3 support	Yes
		Cut through	Yes
		VLAN	Yes
		Jumbo frame support	Max 64 KB
		MAC table	2048
		VLAN table	4096
		Stream (perfect filter, layer 3, layer 2) table	256
		Layer 2/Layer 3 update rule table	256
	gPTP	Number of channels	2
	PTP pulse generator	Number of channels	4
Ethernet AVB	IEEE802.1BA	The AVB profile	Yes
	IEEE1722-2011	Media transport protocol for audio and video	Yes
	IEEE 802.1AS-2011	gPTP (a profile of IEEE1588 for AVB systems)	Yes
	IEEE 802.1Qav-2009	Credit based shaper for bandwidth reservation	Yes
Ethernet TSN	IEEE802.1AS-Rev	Adds redundancy and enhancements to gPTP	Yes
	IEEE 802.1Qbv-2015	Time aware shaper	Yes
	IEEE 802.1Qbu-2016 (802.3br)	Frame preemption	Yes
	IEEE 802.1Qch-2017	Cyclic queuing and forwarding	Yes
	IEEE 802.1Qcr	Asynchronous traffic shaping	Yes
	IEEE 802.1Qci-2017	Stream based filtering and policing	Yes (without gate filter)
	IEEE 802.1CB-2017	Frame replication and elimination for reliability	Yes
TrustZone Filter	Security attribution and privilege attribution can be set		

Note: Please contact a Renesas Electronics sales office for network support.

**Figure 30.1 ESWM block diagram****Table 30.2 ESWM functional blocks (1 of 2)**

Block name	Function
MFWD	Ethernet Message Forwarding Engine (MFWD) is part of ESWM and aims to filter, forward, and route the switch frames. MFWD snoops the frame information from fabric to redistribute them to the agent after filtering, forwarding, and routing mechanisms.
MFAB	Fabric Bus (MFAB) is an internal bus used to exchange data between Ethernet Agent [ETHA], Ethernet Common Agent [COMA], Ethernet CPU Agent [GWCA], local RAM, TAG RAM, pointer RAM, and Ethernet Message Forwarding Engine [MFWD].
ETHA	Ethernet Agent (ETHA) consists of an agent interface module to allow communication within ESWM. ETHA handles data exchange between ESWM and an Ethernet PHY.
RMAC	Ethernet MAC (RMAC) is an Ethernet controller that conforms to the definition of the MAC (Media Access Control) layer for Ethernet in the IEEE 802.3 standard. By connecting with a physical-layer LSI chip (PHY-LSI) that complies with the standard, RMAC can transmit and receive Ethernet (IEEE 802.3) frames. RMAC has a single MAC layer interface.
GWCA	Ethernet CPU Agent (GWCA) is a CPU interface that consists of an agent interface module allowing communication within ESWM. GWCA handles data exchange between ESWM and the GWCPU subsystem. Software is required for the configuration of Gateway AXI bus interface.
COMA	Ethernet Common Agent (COMA) is an agent of ESWM which aims at gathering common functionalities for other agents. COMA handles the APB to SFR bus conversion, the buffer pointer release for rejected frames and the pointer handling for local RAM. Software is required for the configuration of Ethernet Common Agent.

Table 30.2 ESWM functional blocks (2 of 2)

Block name	Function
gPTP	Ethernet Generic PTP timer (GPTP) enables accurate synchronization of the clock in the control system.
plsotim	The pulse output timer module (plsotim), a submodule of gPTP, produces a pulsed output according to the base period of a gPTP timer.

30.2 Register Descriptions

30.2.1 ESWM Interface Control Registers

30.2.1.1 MIIRR : Media-independent Interface Reset Register

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x19400

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	RMRS T1	RMRS T0	—	—	—	—	—	—	RGRS T1	RGRS T0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RGRST0	RGMII0 Reset 0: Reset 1: Enable	R/W
1	RGRST1	RGMII1 Reset 0: Reset 1: Enable	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
8	RMRST0	RMII0 Reset 0: Reset 1: Enable	R/W
9	RMRST1	RMII1 Reset 0: Reset 1: Enable	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

30.2.1.2 MIICR0 : Media-independent Interface Control Register 0

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x19404

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	TXCIDE	—	—	—	DIVSTP	—	—	—	—	—	—	MIISEL[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	MIISEL[1:0]	Media-independent Interface Select 0 0: MII/GMII 0 1: RGMII 1 0: RMII 1 1: Setting prohibited	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
8	DIVSTP	Clock Divider Stop 0: Enable 1: Stop	R/W
11:9	—	These bits are read as 0. The write value should be 0.	R/W
12	TXCIDE	RGMII TxC Internal Delay Enable 0: TxC internal delay disable 1: TxC internal delay enable	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

30.2.1.3 MIICR1 : Media-independent Interface Control Register 1

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x19408

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	TXCIDE	—	—	—	DIVSTP	—	—	—	—	—	—	MIISEL[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	MIISEL[1:0]	Media-independent Interface Select 0 0: MII/GMII 0 1: RGMII 1 0: RMII 1 1: Setting prohibited	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
8	DIVSTP	Clock Divider Stop 0: Enable 1: Stop	R/W
11:9	—	These bits are read as 0. The write value should be 0.	R/W
12	TXCIDE	RGMII TXC Internal Delay Enable 0: TXC Internal delay disable 1: TXC Internal delay enable	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

30.2.1.4 MCCESR : Media Clock Capture Event Select Register

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x19410

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MCCE S1	MCCE S0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MCCES0	Media Clock 0 Capture Event Select 0: I/O Port 1: ELC	R/W
1	MCCES1	Media Clock 1 Capture Event Select 0: I/O Port 1: ELC	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

30.2.1.5 TASSTSR : TAS Status Monitor Signal Select Register

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x19420

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	MSS3[4:0]				—	—	—	MSS2[4:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	MSS1[4:0]				—	—	—	MSS0[4:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
4:0	MSS0[4:0]	ET_TAS_STA0 Monitor Signal Select Select signal to output ET_TAS_STA0 pin from race_etha0_tas_gate_state[8:0] and race_etha1_tas_gate_state[8:0]	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
12:8	MSS1[4:0]	ET_TAS_STA1 Monitor Signal Select Select signal to output ET_TAS_STA1 pin from race_etha0_tas_gate_state[8:0] and race_etha1_tas_gate_state[8:0]	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
20:16	MSS2[4:0]	ET_TAS_STA2 Monitor Signal Select Select signal to output ET_TAS_STA2 pin from race_etha0_tas_gate_state[8:0] and race_etha1_tas_gate_state[8:0]	R/W
23:21	—	These bits are read as 0. The write value should be 0.	R/W
28:24	MSS3[4:0]	ET_TAS_STA3 Monitor Signal Select Select signal to output ET_TAS_STA3 pin from race_etha0_tas_gate_state[8:0] and race_etha1_tas_gate_state[8:0]	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

30.2.2 Register Descriptions of Related ESWM Interrupts

30.2.2.1 TPEMIMC1 : Error and Monitoring Interrupt Mapping Configuration Register 1

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x0004

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
Bit field:	—	CSICM[2:0]				—	—	CSIG M	CSIM	—	CEICM[2:0]			—	—	CEIG M	CEIM
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Bit field:	—	FSICM[2:0]				—	—	FSIG M	FSIM	—	FEICM[2:0]			—	—	FEIG M	FEIM
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Bit	Symbol	Function	R/W
0	FEIM	MFWD Error Interrupt Mapping MFWD Error Interrupt contains the following interrupts: • FWEIS0 • FWEIS1 • FWEIS2 • FWEIS3 • FWEIS4 • FWEIS5 • FWEIS6 • FWEIS7 • FWEIS8 0: MFWD Error Interrupt is mapped to ETHER_FWEI 1: MFWD Error Interrupt is mapped to a specific core depending on TPEMIMC1.FEIGM register	R/W
1	FEIGM	MFWD Error Interrupt GWCA Mapping 0: When TPEMIMC1.FEIM is set to 1, MFWD Error Interrupt is mapped to ETHER_GWDI[TPEMIMC1.FEICM] 1: Setting prohibited	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
6:4	FEICM[2:0]	MFWD Error Interrupt Core Mapping Determine to which pin MFWD Error Interrupt is mapped when mapped to a GWCA	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
8	FSIM	MFWD Status Interrupt Mapping MFWD Status Interrupt contains the following interrupt: FWMIS0 [MFWD] 0: MFWD Status Interrupt is mapped to ETHER_FWSI 1: MFWD Status Interrupt is mapped to a specific core depending on TPEMIMC1.FSIGM register	R/W
9	FSIGM	MFWD Status Interrupt GWCA Mapping 0: When TPEMIMC1.FSIM is set to 1, MFWD Status Interrupt is mapped to ETHER_GWDI[TPEMIMC1.FSICM] 1: Setting prohibited	R/W
11:10	—	These bits are read as 0. The write value should be 0.	R/W
14:12	FSICM[2:0]	MFWD Status Interrupt Core Mapping Determine to which pin MFWD Status Interrupt is mapped when mapped to a GWCA	R/W
15	—	This bit is read as 0. The write value should be 0.	R/W
16	CEIM	COMA Error Interrupt Mapping COMA Error Interrupt contains the following interrupts: - CAEIS0 [COMA] - CAEIS1 [COMA] 0: COMA Error Interrupt is mapped to ETHER_CAEI 1: COMA Error Interrupt is mapped to a specific core depending on TPEMIMC1.CEIGM register	R/W
17	CEIGM	COMA Error Interrupt GWCA Mapping 0: When TPEMIMC1.CEIM is set to 1, COMA Error Interrupt is mapped to ETHER_GWDI[TPEMIMC1.CEICM] 1: Setting prohibited	R/W
19:18	—	These bits are read as 0. The write value should be 0.	R/W
22:20	CEICM[2:0]	COMA Error Interrupt Core Mapping Determine to which pin COMA Error Interrupt is mapped when mapped to a GWCA	R/W
23	—	This bit is read as 0. The write value should be 0.	R/W
24	CSIM	COMA Status Interrupt Mapping COMA Status Interrupt contains the following interrupts: - CAMIS0 [COMA] - CAMIS1 [COMA] 0: COMA Status Interrupt is mapped to ETHER_CAMI 1: COMA Status Interrupt is mapped to a specific core depending on TPEMIMC1.CSIGM register	R/W
25	CSIGM	COMA Status Interrupt GWCA Mapping 0: When TPEMIMC1.CSIM is set to 1, COMA Status Interrupt is mapped to ETHER_GWDI[TPEMIMC1.CSICM] 1: Setting prohibited	R/W
27:26	—	These bits are read as 0. The write value should be 0.	R/W
30:28	CSICM[2:0]	COMA Status Interrupt Core Mapping Used to decide to which pin COMA Status Interrupt are mapped when mapped to a GWCA.	R/W
31	—	This bit is read as 0. The write value should be 0.	R/W

30.2.2.2 TPEMIMC2 : Error and Monitoring Interrupt Mapping Configuration Register 2

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x0008

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	GEICM0[2:0]			—	—	GEIGM0	GEIM0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	GEIM0	GWCA0 Error Interrupt Mapping GWCA0 Error Interrupt contains the following interrupts: • GWEIS0 except GWEIS0.TDFES for GWCA0 [GWCA] • GWEIS1 for GWCA0 [GWCA] • GWEIS4 for GWCA0 [GWCA] • GWEIS5 for GWCA0 [GWCA] 0: GWCA0 Error Interrupt is mapped to ETHER_GWEI 1: GWCA0 Error Interrupt is mapped to a specific core depending on TPEMIMC2.GEIGM0 register	R/W
1	GEIGM0	GWCA0 Error Interrupt GWCA Mapping 0: When TPEMIMC2.GEIM0 is set to 1, GWCA0 Error Interrupt is mapped to ETHER_GWDI[TPEMIMC2.GEICM0] 1: Setting prohibited	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
6:4	GEICM0[2:0]	GWCA0 Error Interrupt Core Mapping Determine to which pin GWCA0 Error Interrupt is mapped when mapped to a GWCA.	R/W
31:7	—	These bits are read as 0. The write value should be 0.	R/W

30.2.2.3 TPEMIMC3 : Error and Monitoring Interrupt Mapping Configuration Register 3

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x000C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	ESICM0[2:0]			—	—	ESIGM0	ESIM0	—	EEICM0[2:0]			—	—	EEIGM0	EEIM0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	EEIM0	ETHA0 Error Interrupt Mapping ETHA0 Error Interrupt contains the following interrupts: - EAEIS0 for ETHA0 [ETHA] - EAEIS1 for ETHA0 [ETHA] - EAEIS2 for ETHA0 [ETHA] - MEIS for RMAC0 [RMAC] 0: ETHA0 Error Interrupt is mapped to ETHER_EAEI0 1: ETHA0 Error Interrupt is mapped to a specific core depending on TPEMIMC3.EEIGM0 register	R/W
1	EEIGM0	ETHA0 Error Interrupt GWCA Mapping 0: When TPEMIMC3.EEIM0 is set to 1, ETHA0 Error Interrupt is mapped to ETHER_GWDI[TPEMIMC3.EEICM0] 1: Setting prohibited	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
6:4	EEICM0[2:0]	ETHA0 Error Interrupt Core Mapping Determine to which pin ETHA0 Error Interrupt is mapped when mapped to a GWCA.	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
8	ESIM0	ETHA0 Status Interrupt Mapping ETHA0 Status Interrupt contains the following interrupt: MMIS0 for RMAC 0 [RMAC] 0: ETHA0 Status Interrupt is mapped to ETHER_EASI0 1: ETHA0 Status Interrupt is mapped to a specific core depending on TPEMIMC3.ESIGM0 register	R/W
9	ESIGM0	ETHA0 Status Interrupt GWCA Mapping 0: When TPEMIMC3.ESIM0 is set to 1, ETHA0 Status Interrupt is mapped to ETHER_GWDI[TPEMIMC3.ESICM0] 1: Setting prohibited	R/W
11:10	—	These bits are read as 0. The write value should be 0.	R/W
14:12	ESICM0[2:0]	ETHA0 Status Interrupt Core Mapping Determine to which pin ETHA0 Status Interrupt is mapped when mapped to a GWCA.	R/W
31:15	—	These bits are read as 0. The write value should be 0.	R/W

30.2.2.4 TPEMIMC4 : Error and Monitoring Interrupt Mapping Configuration Register 4

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x0010

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	ESICM1[2:0]			—	—	ESIGM1	ESIM1	—	EEICM1[2:0]			—	—	EEIGM1	EEIM1
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	EEIM1	ETHA1 Error Interrupt Mapping ETHA1 Error Interrupt contains the following interrupt: - EAEIS0 for ETHA1 [ETHA] - EAEIS1 for ETHA1 [ETHA] - EAEIS2 for ETHA1 [ETHA] - MEIS for RMAC1 [RMAC] 0: ETHA1 Error Interrupt is mapped to ETHER_EAEI1 1: ETHA1 Error Interrupt is mapped to a specific core depending on TPEMIMC4.EEIGM1 register	R/W

Bit	Symbol	Function	R/W
1	EEIGM1	ETHA1 Error Interrupt GWCA Mapping 0: When TPEMIMC4.EEIM1 is set to 1, ETHA1 Error Interrupt is mapped to ETHER_GWDI[TPEMIMC4.EEICM1] 1: Setting prohibited	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
6:4	EEICM1[2:0]	ETHA1 Error Interrupt Core Mapping Determine to which pin ETHA1 Error Interrupt is mapped when mapped to a GWCA.	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
8	ESIM1	ETHA1 Status Interrupt Mapping ETHA1 Status Interrupt contains the following interrupt: • MMIS0 for RMAC 1 [RMAC] 0: ETHA1 Status Interrupt is mapped to ETHER_EAS1 1: ETHA1 Status Interrupt is mapped to a specific core depending on TPEMIMC4.ESIGM1 register	R/W
9	ESIGM1	ETHA1 Status Interrupt GWCA Mapping 0: When TPEMIMC4.ESIM1 is set to 1, ETHA1 Status Interrupt is mapped to ETHER_GWDI[TPEMIMC4.ESICM1] 1: Setting prohibited	R/W
11:10	—	These bits are read as 0. The write value should be 0.	R/W
14:12	ESICM1[2:0]	ETHA1 Status Interrupt Core Mapping Determine to which pin ETHA1 Status Interrupt is mapped when mapped to a GWCA.	R/W
31:15	—	These bits are read as 0. The write value should be 0.	R/W

30.2.2.5 TPEMIMC6t : Error and Monitoring Interrupt Mapping Configuration Register 6t (t = 0 to 1)

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x0080 + 0x04 × t

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	GTSICM0[2:0]		GTSIM0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	GTSIM0	GWCA0 Timestamp Interrupt Mapping GWCA0 Timestamp Interrupt contains the following interrupts: • GWTDIS.TSDIS for GWCA0 [GWCA] • GWEIS0.TDFES for GWCA0 [GWCA] 0: GWCA0 Timestamp Interrupt t is mapped to ETHER_TSDI[t] 1: GWCA0 Timestamp Interrupt t is mapped to ETHER_GWDIn depending on TPEMIMC6t.GTSICM0 setting	R/W
3:1	GTSICM0[2:0]	GWCA0 Timestamp Interrupt Core Mapping Map GWCA0 Timestamp Interrupt t to ETHER_GWDI[TPEMIMC6t.GTSICM0] when TPEMIMC6t.GTSIM0 is set.	R/W
31:4	—	These bits are read as 0. The write value should be 0.	R/W

30.2.2.6 TPEMIMC7t : Error and Monitoring Interrupt Mapping Configuration Register 7t (t = 0 to 63)

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x0100 + 0x04 × t

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	GDICM0[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	GDICM0[2:0]	GWCA0 Data Interrupt Core Mapping Map GWCA0 Data Interrupts t to ETHER_GWDI[TPEMIMC7t.GDICM0]. RMAC2 Status Interrupt contains the following interrupts: • GWDISi.DIS for GWCA0 [GWCA] • GWEIS2i.DFES for GWCA0 [GWCA] • GWEIS3.IAOEst for t = 0 to 4 for GWCA0 [GWCA]	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

30.2.2.7 TSIM : Summarized Interrupt Mirroring Register

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x0700

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	EIM1	EIM0	—	GIM0	CIM	FIM
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	FIM	MFWD Interrupt Mirroring 0: No interrupt is set in MFWD 1: An interrupt is set in MFWD	R
1	CIM	COMA Interrupt Mirroring 0: No interrupt is set in COMA agent 1: An interrupt is set in COMA agent	R
2	GIM0	GWCA0 Interrupt Monitoring 0: No interrupt is set in GWCA0 1: An interrupt is set in GWCA0	R
3	—	This bit is read as 0.	R
4	EIM0	ETHA0 Interrupt Monitoring 0: No interrupt is set in ETHA0 1: An interrupt is set in ETHA0	R

Bit	Symbol	Function	R/W
5	EIM1	ETHA1 Interrupt Monitoring 0: No interrupt is set in ETHA1 1: An interrupt is set in ETHA1	R
31:6	—	These bits are read as 0.	R

30.2.2.8 TFIM : MFWD Interrupt Mirroring Register

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x0704

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	FWMISIM0	FWEISIM8	FWEISIM7	FWEISIM6	FWEISIM5	FWEISIM4	FWEISIM3	FWEISIM2	FWEISIM1	FWEISIM0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	FWEISIM0	FWEIS0 Interrupt Mirroring 0: No interrupt is set in forwarding engine FWEIS0 register 1: An interrupt is set in forwarding engine FWEIS0 register	R
1	FWEISIM1	FWEIS1 Interrupt Mirroring 0: No interrupt is set in forwarding engine FWEIS1 register 1: An interrupt is set in forwarding engine FWEIS1 register	R
2	FWEISIM2	FWEIS2 Interrupt Mirroring 0: No interrupt is set in forwarding engine FWEIS2 register 1: An interrupt is set in forwarding engine FWEIS2 register	R
3	FWEISIM3	FWEIS3 Interrupt Mirroring 0: No interrupt is set in forwarding engine FWEIS3 register 1: An interrupt is set in forwarding engine FWEIS3 register	R
4	FWEISIM4	FWEIS4 Interrupt Mirroring 0: No interrupt is set in forwarding engine FWEIS4 register 1: An interrupt is set in forwarding engine FWEIS4 register	R
5	FWEISIM5	FWEIS5 Interrupt Mirroring 0: No interrupt is set in forwarding engine FWEIS5 register 1: An interrupt is set in forwarding engine FWEIS5 register	R
6	FWEISIM6	FWEIS6 Interrupt Mirroring 0: No interrupt is set in forwarding engine FWEIS6 register 1: An interrupt is set in forwarding engine FWEIS6 register	R
7	FWEISIM7	FWEIS7 Interrupt Mirroring 0: No interrupt is set in forwarding engine FWEIS7 register 1: An interrupt is set in forwarding engine FWEIS7 register	R
8	FWEISIM8	FWEIS8 Interrupt Mirroring 0: No interrupt is set in forwarding engine FWEIS8 register 1: An interrupt is set in forwarding engine FWEIS8 register	R
9	FWMISIM0	FWMIS0 Interrupt Mirroring 0: No interrupt is set in forwarding engine FWMIS0 register 1: An interrupt is set in forwarding engine FWMIS0 register	R
31:10	—	These bits are read as 0.	R

30.2.2.9 TCIM : COMA Interrupt Mirroring Register

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x0708

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	CAMIS IM1	CAMIS IM0	CAEIS IM1	CAEIS IM0	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0.	R
1	CAEISIM0	CAEIS0 Interrupt Mirroring 0: No interrupt is set in COMA agent CAEIS0 register 1: An interrupt is set in COMA agent CAEIS0 register	R
2	CAEISIM1	CAEIS1 Interrupt Mirroring 0: No interrupt is set in COMA agent CAEIS1 register 1: An interrupt is set in COMA agent CAEIS1 register	R
3	CAMISIM0	CAMIS0 Interrupt Mirroring 0: No interrupt is set in COMA agent CAMIS0 register 1: An interrupt is set in COMA agent CAMIS0 register	R
4	CAMISIM1	CAMIS1 Interrupt Mirroring 0: No interrupt is set in COMA agent CAMIS1 register 1: An interrupt is set in COMA agent CAMIS1 register	R
31:5	—	These bits are read as 0.	R

30.2.2.10 TGIM0 : GWCA0 Interrupt Mirroring Register

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x0710

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	GWEI SIM5	GWEI SIM4	GWEI SIM3	GWEI SIM2	GWEI SIM1	GWEI SIM0	GWTS DISIM	GWDI SIM
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	GWDISIM	GWDIS Interrupt Mirroring 0: No interrupt is set in GWCA0 GWDIS register 1: An interrupt is set in GWCA0 GWDIS register	R
1	GWTS DISIM	GWTS DISIM Interrupt Mirroring 0: No interrupt is set in GWCA0 GWTS DISIM register 1: An interrupt is set in GWCA0 GWTS DISIM register	R

Bit	Symbol	Function	R/W
2	GWEISIM0	GWEIS0 Interrupt Mirroring 0: No interrupt is set in GWCA0 GWEIS0 register 1: An interrupt is set in GWCA0 GWEIS0 register	R
3	GWEISIM1	GWEIS1 Interrupt Mirroring 0: No interrupt is set in GWCA0 GWEIS1 register 1: An interrupt is set in GWCA0 GWEIS1 register	R
4	GWEISIM2	GWEIS2 Interrupt Mirroring 0: No interrupt is set in GWCA0 GWEIS2 register 1: An interrupt is set in GWCA0 GWEIS2 register	R
5	GWEISIM3	GWEIS3 Interrupt Mirroring 0: No interrupt is set in GWCA0 GWEIS3 register 1: An interrupt is set in GWCA0 GWEIS3 register	R
6	GWEISIM4	GWEIS4 Interrupt Mirroring 0: No interrupt is set in GWCA0 GWEIS4 register 1: An interrupt is set in GWCA0 GWEIS4 register	R
7	GWEISIM5	GWEIS5 Interrupt Mirroring 0: No interrupt is set in GWCA0 GWEIS5 register. 1: An interrupt is set in GWCA0 GWEIS5 register.	R
31:8	—	These bits are read as 0.	R

30.2.2.11 TEIM0 : ETHA0 Interrupt Mirroring Register

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x0720

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	MMISI M	MEISI M	EAEIS IM2	EAEIS IM1	EAEIS IM0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	EAEISIM0	EAEIS0 Interrupt Mirroring 0: No interrupt is set in ETHA0 EAEIS0 register 1: An interrupt is set in ETHA0 EAEIS0 register	R
1	EAEISIM1	EAEIS1 Interrupt Mirroring 0: No interrupt is set in ETHA0 EAEIS1 register 1: An interrupt is set in ETHA0 EAEIS1 register	R
2	EAEISIM2	EAEIS2 Interrupt Mirroring 0: No interrupt is set in ETHA0 EAEIS2 register 1: An interrupt is set in ETHA0 EAEIS2 register	R
3	MEISIM	MEIS Interrupt Mirroring 0: No interrupt is set in ETHA0 RMAC MEIS register 1: An interrupt is set in ETHA0 RMAC MEIS register	R
4	MMISIM	MMIS0 Interrupt Mirroring 0: No interrupt is set in ETHA0 RMAC MMIS0 register 1: An interrupt is set in ETHA0 RMAC MMIS0 register	R
31:5	—	These bits are read as 0.	R

30.2.2.12 TEIM1 : ETHA1 Interrupt Mirroring Register

Base address: ESWM = 0x403C_8000
ESWM_NS = 0x503C_8000

Offset address: 0x0724

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	MMISI M	MEISI M	EAEIS IM2	EAEIS IM1	EAEIS IM0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	EAEISIM0	EAEIS0 Interrupt Mirroring 0: No interrupt is set in ETHA1 EAEIS0 register 1: An interrupt is set in ETHA1 EAEIS0 register	R
1	EAEISIM1	EAEIS1 Interrupt Mirroring 0: No interrupt is set in ETHA1 EAEIS1 register 1: An interrupt is set in ETHA1 EAEIS1 register	R
2	EAEISIM2	EAEIS2 Interrupt Mirroring 0: No interrupt is set in ETHA1 EAEIS2 register 1: An interrupt is set in ETHA1 EAEIS2 register	R
3	MEISIM	MEIS Interrupt Mirroring 0: No interrupt is set in ETHA1 RMAC MEIS register 1: An interrupt is set in ETHA1 RMAC MEIS register	R
4	MMISIM	MMIS0 Interrupt Mirroring 0: No interrupt is set in ETHA1 RMAC MMIS0 register 1: An interrupt is set in ETHA1 RMAC MMIS0 register	R
31:5	—	These bits are read as 0.	R

30.2.3 ESWM Core Registers

See the following sections:

- [section 31.3. MFWD Register Descriptions](#)
- [section 32.3. COMA Register Descriptions](#)
- [section 33.3. ETHA Register Descriptions](#)
- [section 34.4. RMAC Register Descriptions](#)
- [section 35.3. GWCA Register Descriptions](#)

30.2.4 gPTP and PTP Pulse Generator Registers

See the following sections:

- [section 36.3. gPTP Register Descriptions](#)
- [section 36.6.3. Plsotim Register Descriptions](#)

30.3 Register Utilization

30.3.1 Software Flows

Restrictions on software flows are as follows:

- SW: ESWM registers can only be written using the flows specified in this section

- SW: All the software flows described in this section can only be used sequentially.

30.3.1.1 Software Flow Legend

Figure 30.2 shows the software flow legend.

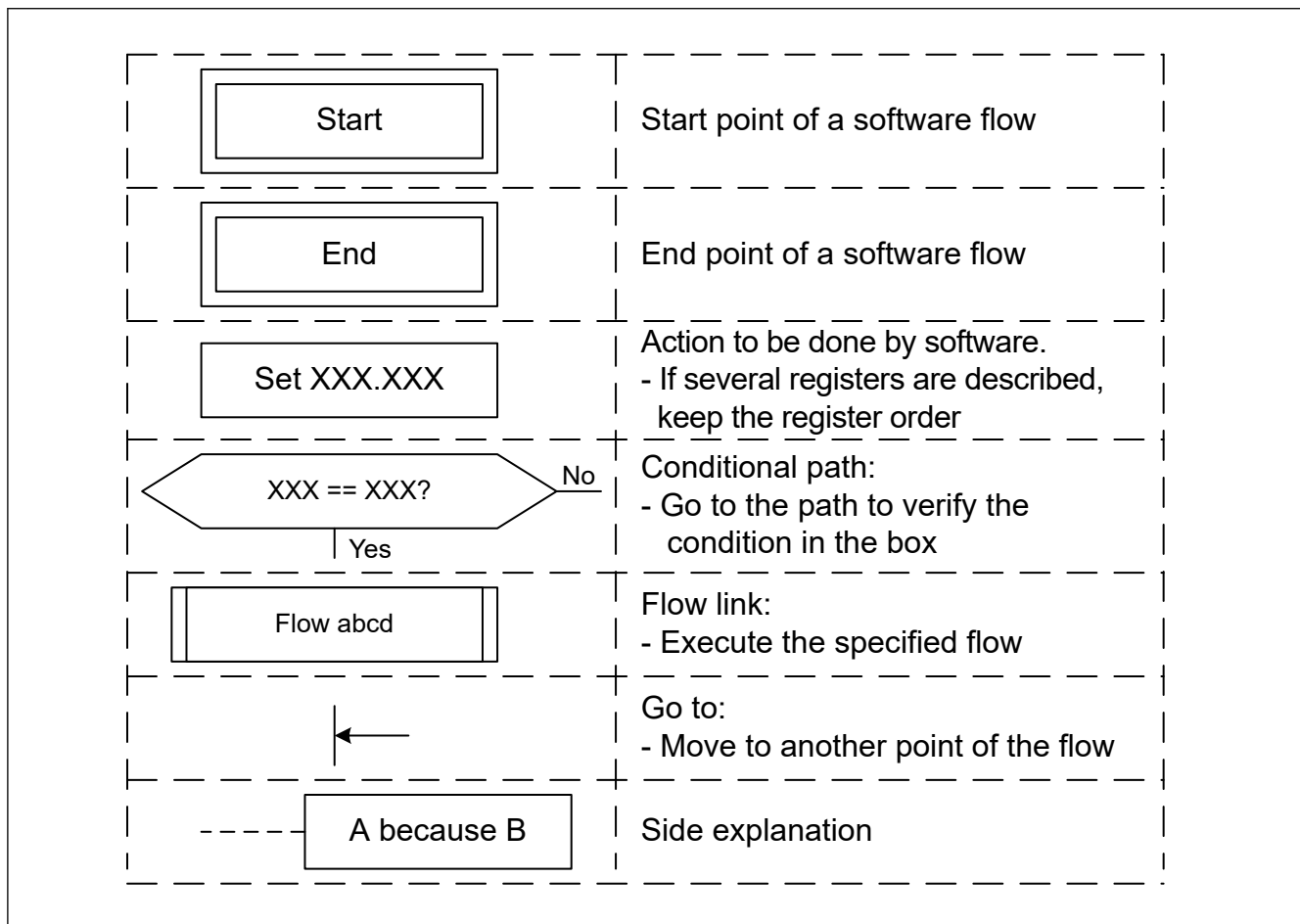


Figure 30.2 Software flow legend

30.3.1.2 Switch Initialization Flow

Figure 30.3 shows the switch initialization flow.

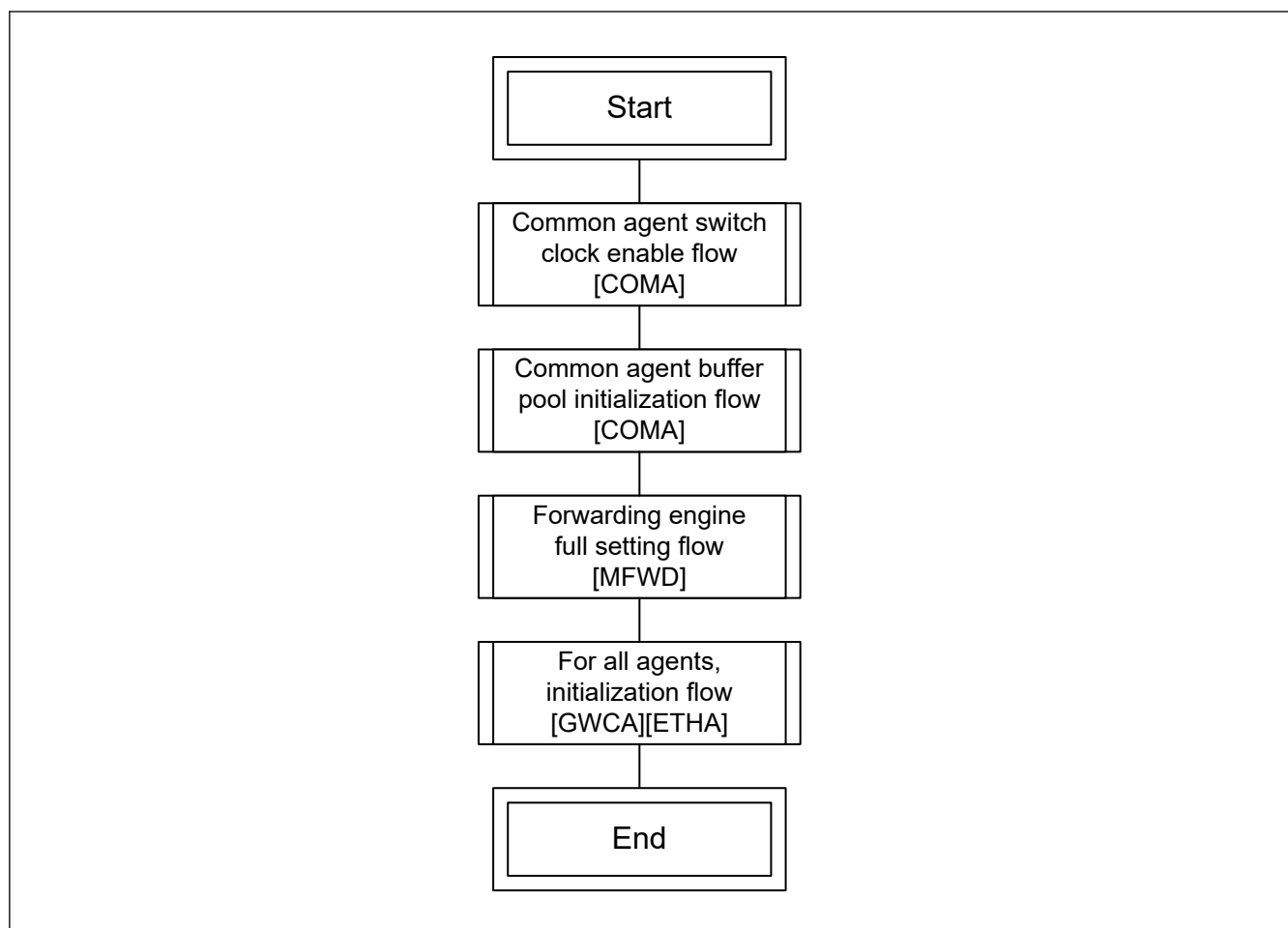


Figure 30.3 Switch initialization flow

30.3.1.3 Switch Disable Flow

[Figure 30.4](#) shows the switch disable flow.

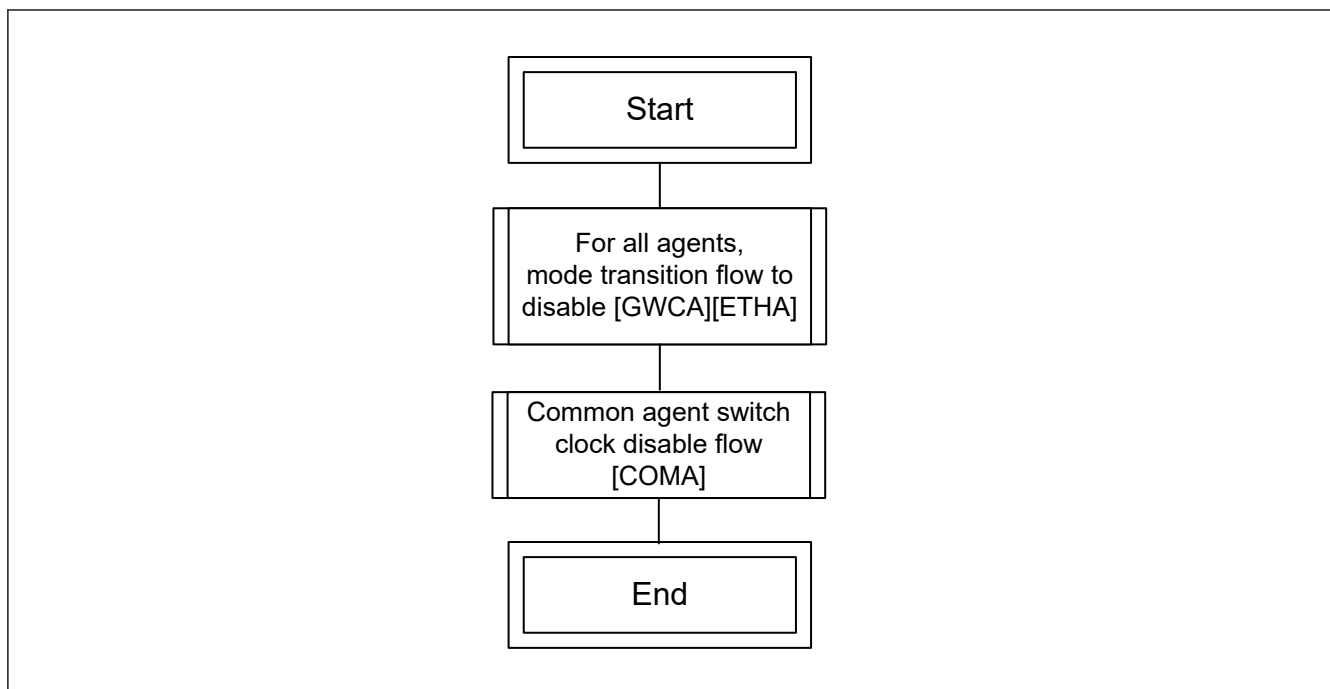


Figure 30.4 Switch disable flow

30.3.1.4 Switch Reset Flow

Figure 30.5 shows the switch reset flow.

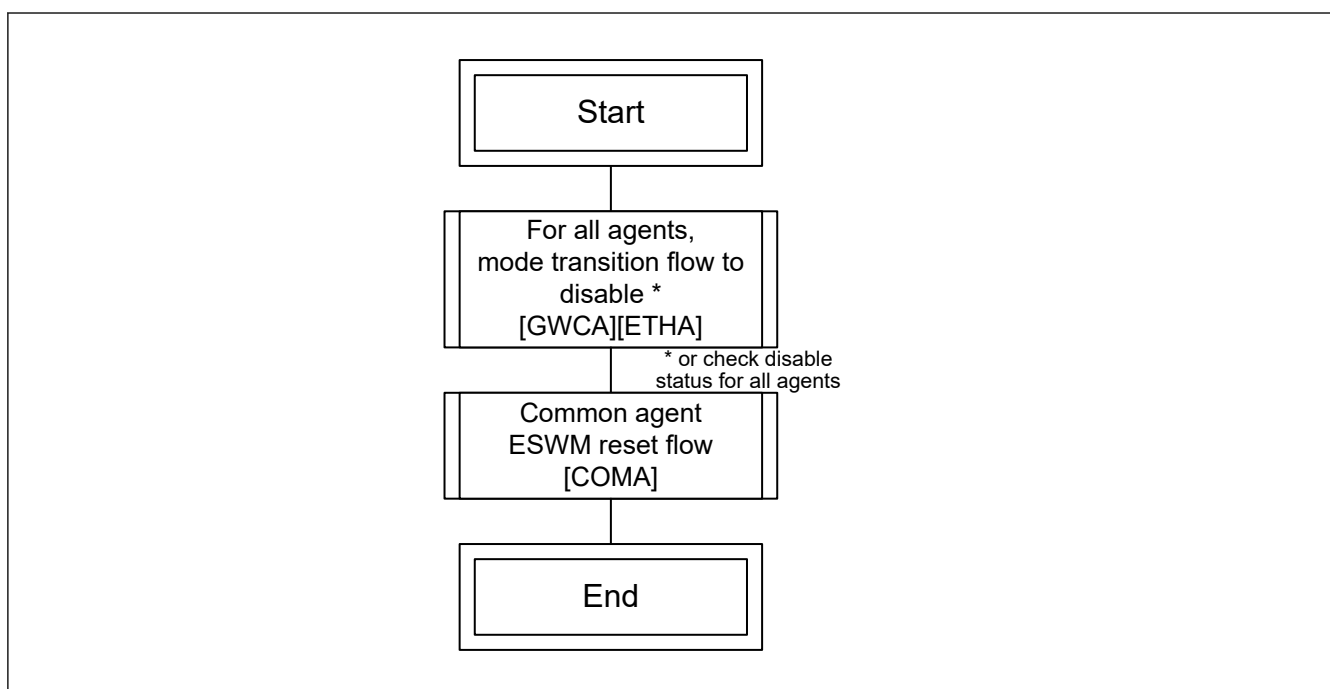


Figure 30.5 Switch reset flow

30.3.1.5 Switch Emergency Reset Flow

Figure 30.6 shows the switch emergency reset flow used when [ETHA] is locked.

For example, in the case of MII, if an error occurs in the PHY and the MII PHY TX clock stops, the switch cannot send data and cannot release the pointer (locked on [ETHA] Operation mode). The state clears when the MII PHY TX clock is restarted, but use emergency reset if required.

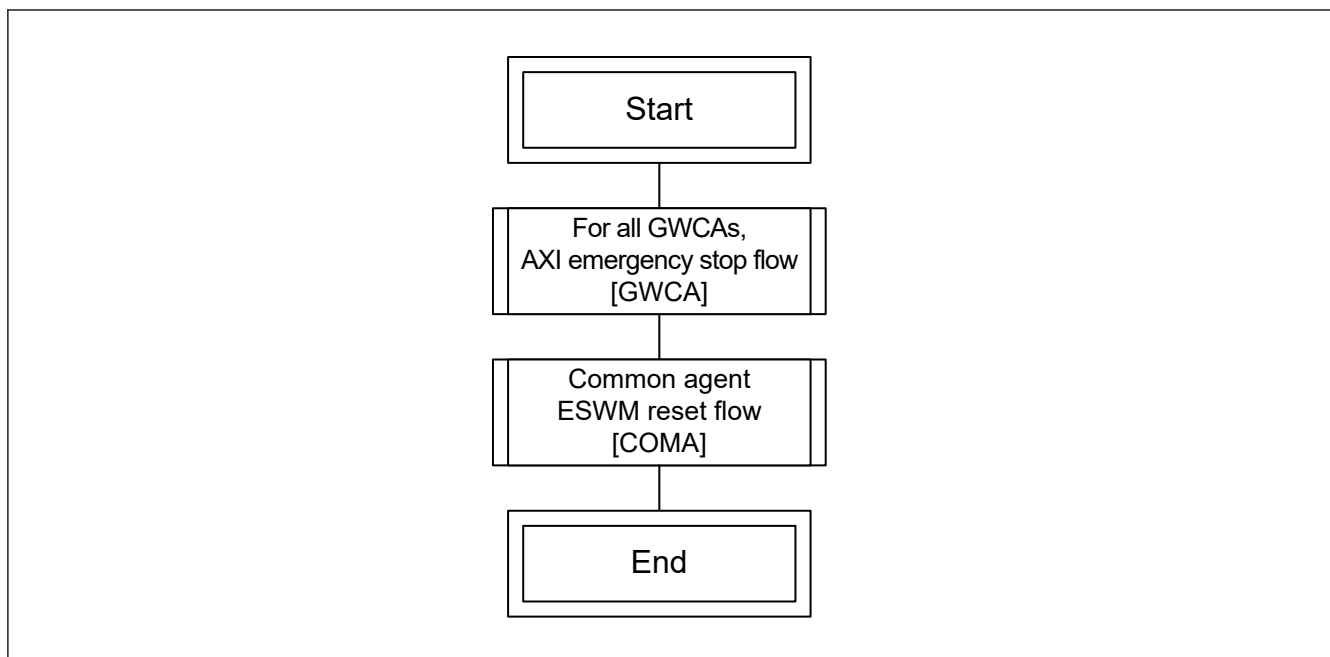


Figure 30.6 Switch emergency reset flow

30.3.1.6 Register Writable Without Software Flow

Table 30.3 describes the registers writable without software flow and in which mode they can be written.

Table 30.3 Writable without software flow registers (✓ = writable, × = not writable)

Register name	Writable anytime
TPEMIMC0	✓
TPEMIMC1	✓
TPEMIMC2	✓
TPEMIMC3	✓
TPEMIMC4	✓
TPEMIMC5	✓
TPEMIMC6t	✓
TPEMIMC7t	✓
TPICVC0	✓
TPICVC1	✓
TPICVC2	✓
TPRCVC	✓
TCVUC	✓

30.4 Functional Descriptions

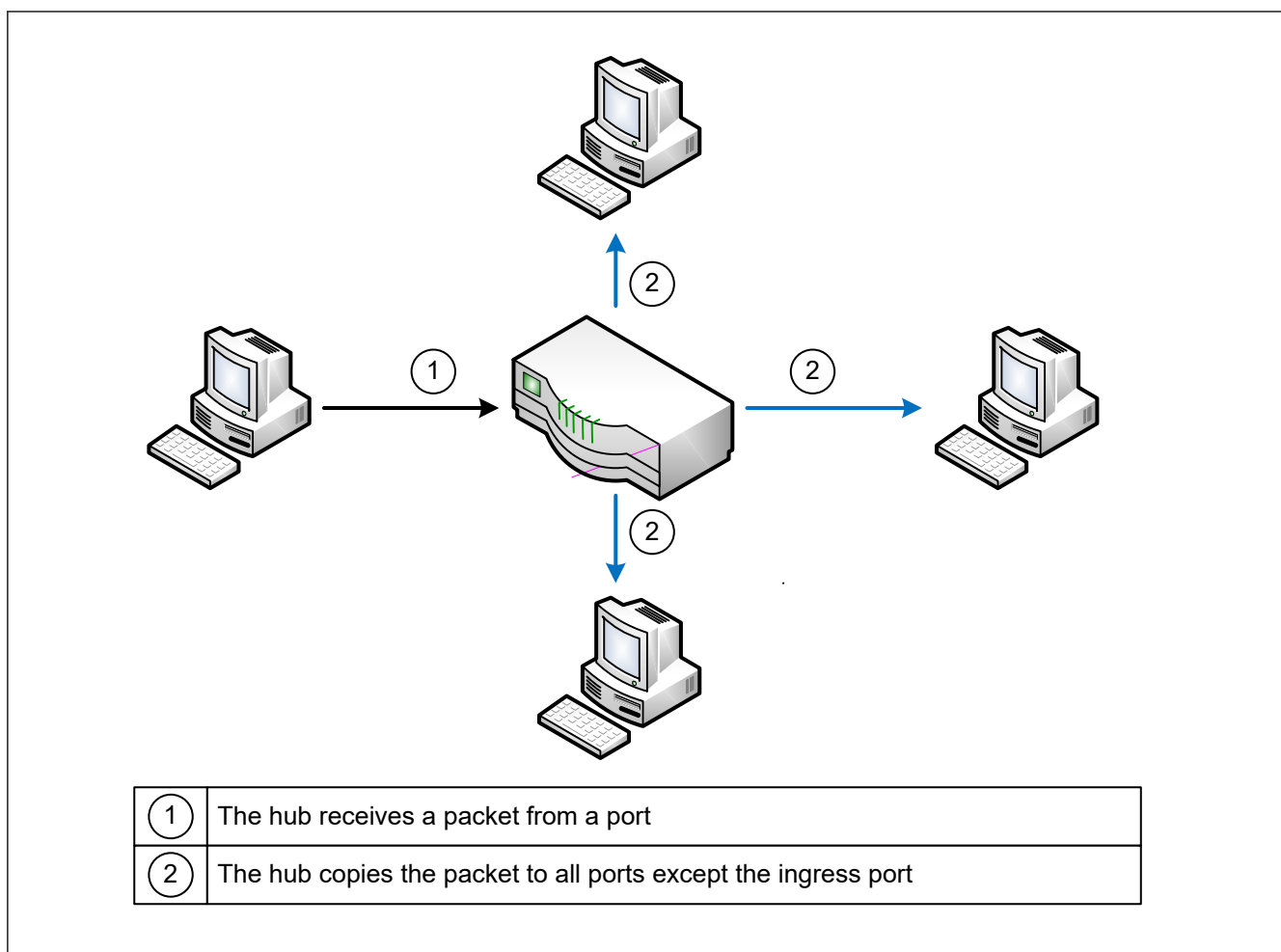
30.4.1 ESWM Forwarding Modes

30.4.1.1 Hub

30.4.1.1.1 Background

A hub is an ethernet network interconnection system (OSI) which only works on layer 1. It does not consider any kind of addressing and, consequently, is not aware of the kind of traffic that goes through it and can create security issues.

Figure 30.7 shows the hub operation.

**Figure 30.7 Hub operation**

Note: In Hub mode, the internal port [GWCA] behaves like an ethernet port. In the following configuration, the internal port receives all frames by AXI descriptor queue number 0 and transmits all frames by AXI descriptor queue number 1.

30.4.1.1.2 Settings

Table 30.4 describes the setting required for ESWM in Hub mode. Specify this setting using the switch initialization flow described in [section 30.3.1.2. Switch Initialization Flow](#). Any register not described in the table should be kept to its initial value, so it does not need to be set. After using the switch initialization flow, any used port should be moved to Operation mode when required.

Table 30.4 Hub settings (1 of 2)

Register name	Setting value	Description
Message Forwarding Engine [MFWD]		
FWPBFCi.PBDV	All 1s except for FWPBFCi.PBDV which is set to 0	All ports are set to send their incoming frames to all other ports
Ethernet Agent [ETHA]		
EAIRC	0	All frames are directed to descriptor queue 0
EATDQDCq.DQD	If q = 0, set to 512 Else set to 0	There is no QoS, so all frames are stored in priority 0 queue. As a result, all the output descriptor queue memory is reserved for queue 0.
EAVCC.VEM	011b	Allows SC-tagged frames and C-tagged frames to go through transparently
EATTFC.UT	0	Allows any TAG formats to go through even the unsupported ones

Table 30.4 Hub settings (2 of 2)

Register name	Setting value	Description
Ethernet MAC [RMAC]		
MPIC.PIS	User-defined	Set the PHY interface type
MPIC.LSC	User-defined	Set the PHY speed
MRAFC	03C703C7H	Set MAC reception in promiscuous mode
Gateway CPU Agent [GWCA]		
GWIRC	0	All frames are directed to descriptor queue 0
GWRDQDCq.DQD	If q = 0, set to 512 Else set to 0	There is no QoS, so all frames are stored in priority 0 queue. As a result, all the output descriptor queue memory is reserved for queue 0.
GWVCC.VEM	011	Allows SC-tagged frames and C-tagged frames to go through transparently
GWTTC.UT	0	Allows any TAG formats to go through even the unsupported ones
GWDCBAC0/1	User-defined	Set the LINKFIX table base address
GWDCCi	For i = 1: • Set DQT to 1 • Set OSID to user-defined. Else: • Set OSID to user-defined.	Queue 0 is used for data reception and queue 1 is used for data transmission. All other queues are not used.

30.4.1.2 Layer 2 Switch

30.4.1.2.1 Background

A layer 2 switch is an ethernet network interconnection system (OSI) which only works on layer 2 using MAC addresses. MAC addresses are used to filter and forward messages. A layer 2 switch has three main functions:

- Address learning: A switch can learn MAC addresses by looking at received frame source MAC addresses
- Address aging: A switch can suppress a MAC address if a frame from the corresponding source has not been received since a given period (usually 5 minutes)
- Frame forward/filter: A switch can forward or filter a frame based on its destination MAC address.

Figure 30.8 shows the layer 2 switch operation.

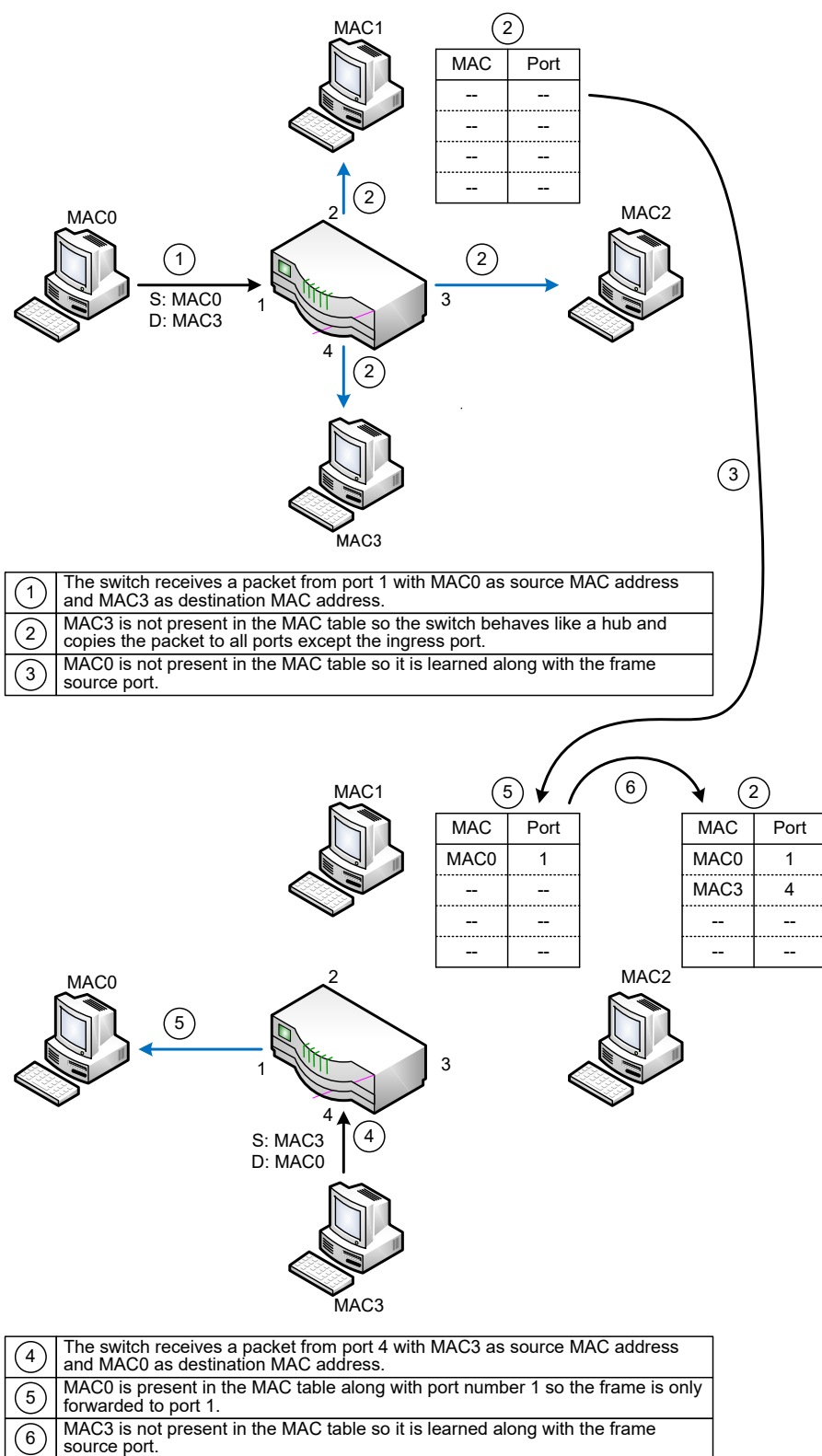


Figure 30.8 Layer 2 switch operation

30.4.1.2.2 Settings

Table 30.5 describes the setting required for ESWM in layer 2 switch mode. Specify this setting using the switch initialization flow described in [section 30.3.1.2. Switch Initialization Flow](#). Any register not described in the table should be kept to its initial value, so it does not need to be set. After using the switch initialization flow, any used port should be moved to Operation mode when required.

Table 30.5 Layer 2 switch settings

Register name	Setting value	Explanation
Message Forwarding Engine [MFWD]		
FWPBFci.PBDV	All 1s except for FWPBFci.PBDV which is set to 0	All ports are set to send their incoming frames to all other ports
FWPCi0.MACDSA	1	MAC destination search occurs for all ports
FWPCi0.MACSSA	1	MAC source search occurs for all ports
FWPCi0.MACHLA	1	MAC source hardware learning is enabled for all ports
FWPCi0.MACHMA	1	MAC source hardware migration is enabled for all ports
FWCLPRC.USMACLF	User-defined	Unknown source MAC addresses can be sent to CPU for learning monitoring
FWCLPTC.LPCSD	2	Set the learning path queue number to 2
FWCLPTC.LPCS	User-defined	Select which CPU to be used for learning monitoring
Ethernet Agent [ETHA]		
EAIRC	0	All frames are directed to descriptor queue 0
EATDQDCq.DQD	If q = 0, set to 512 Else set to 0	There is no QoS, so all frames are stored in priority 0 queue. As a result, all the output descriptor queue memory is reserved for queue 0.
EAVCC.VEM	011b	Allows SC-tagged frames and C-tagged frames to go through transparently
EATTFC.UT	0	Allows any TAG formats to go through even the unsupported ones
Ethernet MAC [RMAC]		
MPIC.PIS	User-defined	Set the PHY interface type
MPIC.LSC	User-defined	Set the PHY speed
MRAFC	0x03C703C7	Set MAC reception in promiscuous mode
Gateway CPU Agent [GWCA]		
GWIRC	0	All frames are directed to descriptor queue 0
GWRDQDCq.DQD	If q = 0, set to 512 Else set to 0	There is no QoS, so all frames are stored in priority 0 queue. As a result, all the output descriptor queue memory is reserved for queue 0.
GWVCC.VEM	011b	Allows SC-tagged frames and C-tagged frames to go through transparently
GWTTC.UT	0	Allows any TAG formats to go through even the unsupported ones
GWDCBAC0/1	User-defined	Set the LINKFIX table base address
GWDCCi	For i = 1: • Set DQT to 1 • Set OSID to user-defined. Else: • Set OSID to user-defined.	Queue 0 is used for data reception, queue 1 is used for data transmission and queue 2 is used for learning monitoring. All other queues are not used. Note: For learning monitoring, use extended descriptors.

Note: Any MAC address learned by hardware coming from an internal port [GWCA] is always learned along with AXI descriptor queue 0. As a result, always reserve AXI descriptor queue 0 for hardware learned MAC addresses that correspond to frames reception queue.

30.4.1.3 VLAN Aware Layer 2 Switch

30.4.1.3.1 Background

A VLAN aware layer 2 switch is an ethernet network interconnection system (OSI) which works on layer 2 using MAC addresses and can reduce its broadcast domains in subdomains using VLAN IDs. A VLAN aware layer 2 switch has the same functions as a layer 2 switch but can also handle VLANs.

[Figure 30.9](#) shows the VLAN aware layer 2 switch operation.

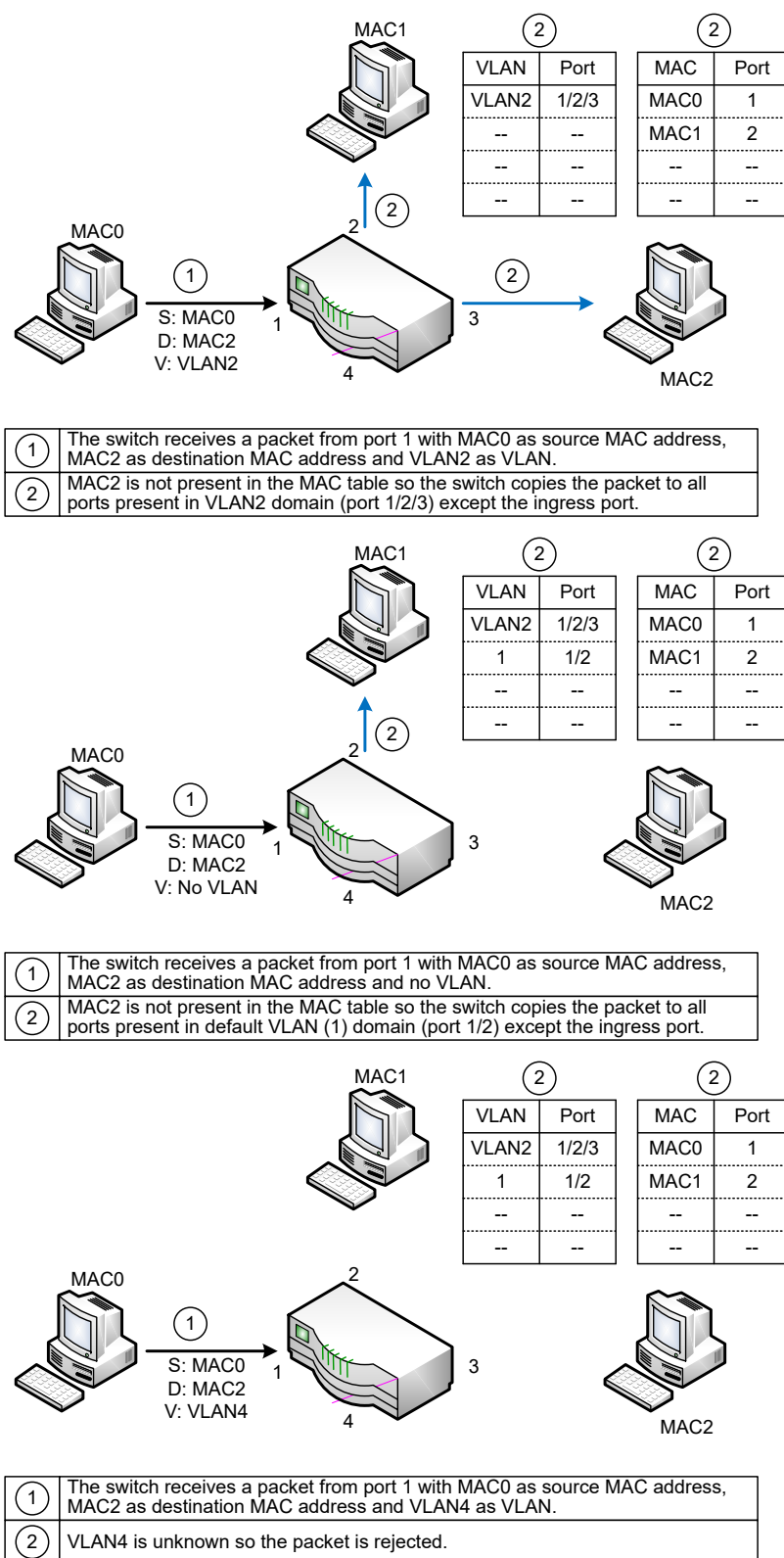


Figure 30.9 VLAN aware layer 2 switch operation

30.4.1.3.2 Settings

Table 30.6 describes the setting required for ESWM in VLAN aware layer 2 switch mode. Specify this setting using the switch initialization flow described in [section 30.3.1.2. Switch Initialization Flow](#). Any register not described in the table should be kept to its initial value, so it does not need to be set. After using the switch initialization flow, any used port should be moved to Operation mode when required.

Table 30.6 VLAN aware layer 2 switch settings

Register name	Setting value	Explanation
Message forwarding engine [MFWD]		
FWGC.SVM	01	Set ESWM in C-TAG mode
FWPBFCi.PBDV	All 1s except for FWPBFCi.PBDV which is set to 0	All ports are set to send their incoming frames to all other ports
FWPCi0.MACDSA	1	MAC destination search occurs for all ports
FWPCi0.MACSSA	1	MAC source search occurs for all ports
FWPCi0.MACHLA	1	MAC source hardware learning is enabled for all ports
FWPCi0.MACHMA	1	MAC source hardware migration is enabled for all ports
FWPCi0.VLANSA	1	VLAN search occurs for all ports
FWPCi0.VLANRU	1	Unknown VLANs are rejected
FWCLPRC.USMACLF	User-defined	Unknown source MAC addresses can be sent to CPU for learning monitoring
FWCLPRC.UVLANLF	User-defined	Unknown VLANs can be sent to CPU for learning
FWCLPTC.LPCSD	2	Set the learning path queue number to 2
FWCLPTC.LPCS	User-defined	Select which CPU to be used for learning monitoring
FWVLANTL0/1/2/30/4	User-defined	New VLANs can be added to the VLAN table. For details, see Table 30.7 .
Ethernet agent [ETHA]		
EATDQDCq.DQD	User-defined	The number of descriptors in descriptor queues can be set
EAVCC.VEM	011	Allows SC-tagged frames and C-tagged frames to go through transparently
EAVTC.CTV	1	Set the default VLAN value to 1
EATTFC.UT	0	Allows any TAG formats to go through even the unsupported ones
RMAC [RMAC]		
MPIC.PIS	User-defined	Set the PHY interface type
MPIC.LSC	User-defined	Set the PHY speed
MRAFC	03C703C7H	Set MAC reception in promiscuous mode
GWCA [GWCA]		
GWRDQDCq.DQD	User-defined	The number of descriptors in descriptor queues can be set
GWVCC.VEM	011	Allows SC-tagged frames and C-tagged frames to go through transparently
GWVTC.CTV	1	Set the default VLAN value to 1
GWTTC.UT	0	Allows any TAG formats to go through even the unsupported ones
GWDCBAC0/1	User-defined	Set the LINKFIX table base address
GWDCCi	For i = 1: • Set DQT to 1 • Set OSID to user-defined. Else: • Set OSID to user-defined.	Queue 0 is used for data reception, queue 1 is used for data transmission and queue 2 is used for learning monitoring. All other queues are not used. Note: For learning monitoring, use extended descriptors.

Table 30.7 VLAN learning in VLAN table

Field name VLAN	Setting value	Explanation
SL	0	No secure entry used
HLD	0	Hardware learning for MAC addresses not disabled for any VLAN
SLV	Set to 1 for ports included in VLAN	Only authorize to receive corresponding VLAN from ports included in this VLAN
DV	Set to 1 for ports included in VLAN	Only authorize to send corresponding VLAN to ports included in this VLAN
CSD	0	Frames received by CPUs on sub-destination 0
CME	0	No mirroring
EME	0	No mirroring
IPU	0	No frame priority update
IPV	0	No frame priority update

Note: When both VLAN and destination MAC addresses are found in forwarding tables and corresponding frame is forwarded to an internal port, the AXI descriptor queue (CPU sub-destination) chosen for forwarding is the one set in the MAC table. In this example, because both are set to 0, it raises no issue but, when AXI descriptor queues are used for virtualization, it should be taken in account during system design.

30.4.1.4 Layer 3 Switch

30.4.1.4.1 Background

A layer 3 switch is an ethernet network interconnection system (OSI) which works on layer 3 using IP addresses. Because ESWM can only perform perfect matches by stream forwarding, when developing a layer 3 switch, it is recommended to use perfect filters [FWD].

Figure 30.10 shows the layer 3 switch operation.

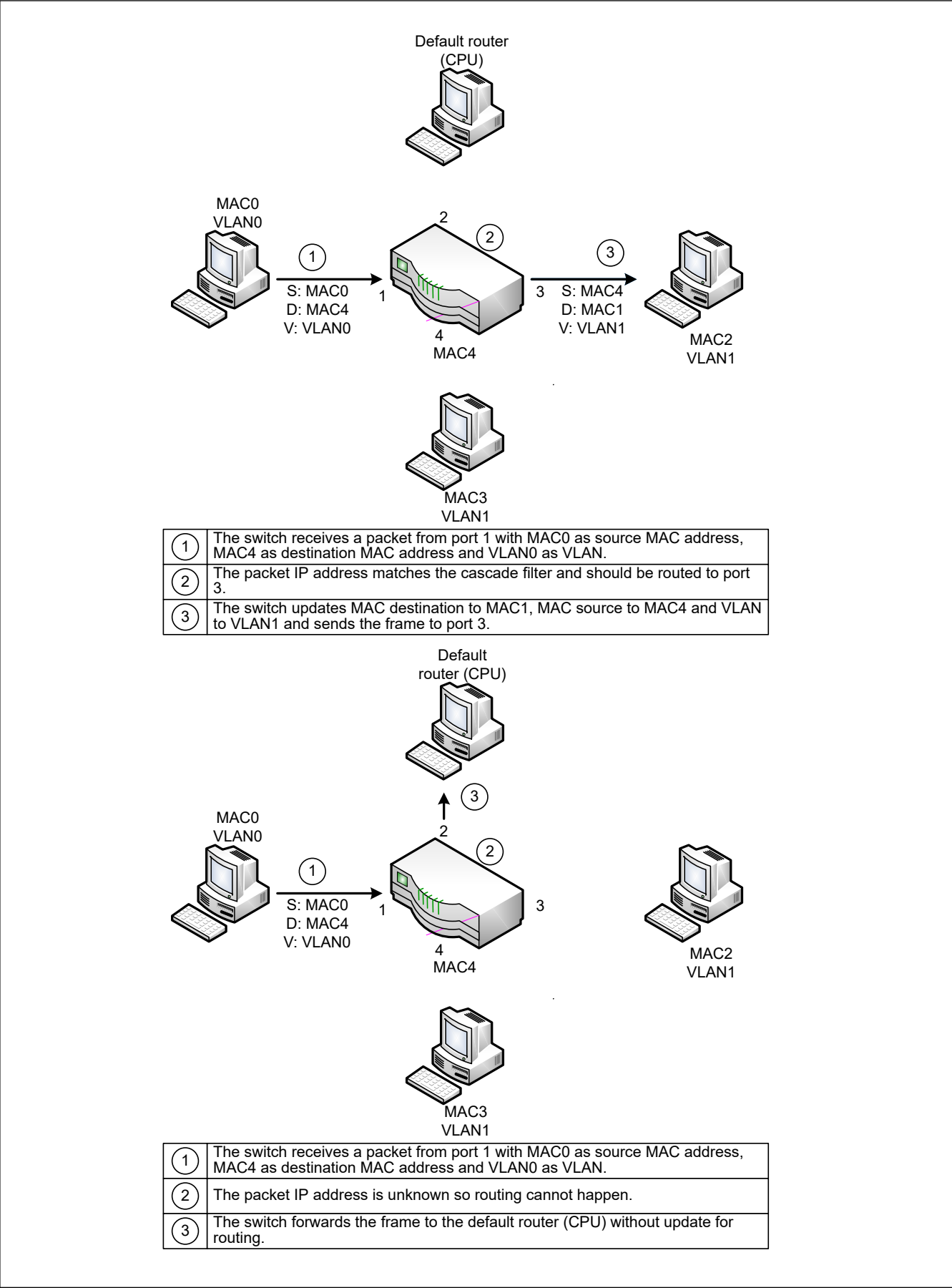


Figure 30.10 Layer 3 switch operation

30.4.1.4.2 Settings

Table 30.8 describes the setting required for ESWM in layer 3 switch mode. Specify this setting using the switch initialization flow described in [section 30.3.1.2. Switch Initialization Flow](#). Any register not described in the table should be kept to its initial value, so it does not need to be set. After using the switch initialization flow, any used port should be moved to Operation mode when required.

Table 30.8 Layer 3 switch settings

Register name	Setting value	Explanation
Forwarding engine [MFWD]		
FWGC.SVM	01	Set ESWM in C-TAG mode
FWPBFCi.PBDV	All 1s except for FWPBFCi.PBDV which is set to 0	All ports are set to send their incoming frames to all other ports
FWPCi0.MACDSA	1	MAC destination search occurs for all ports
FWPCi0.MACSSA	1	MAC source search occurs for all ports
FWPCi0.MACHLA	1	MAC source hardware learning is enabled for all ports
FWPCi0.MACHMA	1	MAC source hardware migration is enabled for all ports
FWPCi0.VLANSA	1	VLAN search occurs for all ports
FWCLPRC.USMACLF	User-defined	Unknown source MAC addresses can be sent to CPU for learning monitoring
FWCLPRC.UVLANLF	1	Unknown VLANs are sent to CPU for learning
FWCLPTC.LPCSD	2	Set the learning path queue number to 2
FWCLPTC.LPCS	User-defined	Select which CPU to be used for learning monitoring
FWVLANTL0/1/2/30/4	User-defined	New VLANs can be added to the VLAN table. For details, see Table 30.7 .
Ethernet agent [ETHA]		
EATDQDCq.DQD	User-defined	The number of descriptors in descriptor queues can be set
EAVCC.VEM	011	Allows SC-tagged frames and C-tagged frames to go through transparently
EAVTC.CTV	1	Set the default VLAN value to 1
EATTFC.UT	0	Allows any TAG formats to go through even the unsupported ones
RMAC [RMAC]		
MPIC.PIS	User-defined	Set the PHY interface type
MPIC.LSC	User-defined	Set the PHY speed
MRAFC	03C703C7H	Set MAC reception in promiscuous mode
GWCA [GWCA]		
GWRDQDCq.DQD	User-defined	The number of descriptors in descriptor queues can be set
GWVCC.VEM	011	Allows SC-tagged frames and C-tagged frames to go through transparently
GWVTC.CTV	1	Set the default VLAN value to 1
GWTTC.UT	0	Allows any TAG formats to go through even the unsupported ones
GWDCBAC0/1	User-defined	Set the LINKFIX table base address
GWDCCi	For i = 1: • Set DQT to 1 • Set OSID to user-defined. Else: • Set OSID to user-defined.	Queue 0 is used for data reception, queue 1 is used for data transmission and queue 2 is used for learning monitoring. All other queues are not used. Note: For learning monitoring, use extended descriptors.

Table 30.9 VLAN learning in VLAN table

Field name VLAN	Setting value	Explanation
SL	0	No secure entry used
HLD	0	Hardware learning for MAC addresses not disabled for any VLAN
SLV	Set to 1 for ports included in VLAN	Only authorize to receive corresponding VLAN from ports included in this VLAN
DV	Set to 1 for ports included in VLAN	Only authorize to send corresponding VLAN to ports included in this VLAN
CSD	0	Frames received by CPUs on sub-destination 0
CME	0	No mirroring
EME	0	No mirroring
IPU	0	No frame priority update
IPV	0	No frame priority update

30.4.1.5 AVB scheduling

30.4.1.5.1 Forwarding and Queuing Enhancements for Time-Sensitive Streams [802.1Qav]

30.4.2 Precautions

- ESWM can support max $(([\text{Unused pointer number}] \times 128) / 4)$ bytes frame (supported maximum frame size). A buffer overflow occurs when the transmit frame size exceeds the supported maximum size. $[\text{Unused pointer number}] = 512 - (\text{Activated port number} \times 6)$.
 - Set to [RMAC] MRFSC.EMXS smaller than maximum frame size for Ethernet ports
 - Do not transmit larger than maximum frame size for CPU ports.
- GWCA can only transmit/receive frames with maximum size of 63488 bytes for the first frame. For the upcoming frames, the maximum frame size can be 65535 bytes provided LINK/LINKFIX descriptors were not used in between the queue.
- When using Cut through, use it so as not to cause a buffer overflow. The following usage is recommended:
 - Only use from-one-to-one communication.
(Example of prohibition : Multicast forwarding, to Ethernet mirroring, Loopback forwarding).
 - Prohibition of the function to stop transmission. Use flow control on reception port.
(Example of prohibition : TX queue pause, CBS, TAS).
- The overall specifications of ESWM are determined on the frequency of the operating clock.
See the following table for details to select the operating clock.

Selection			Definition	
Local RAM size*1	Clk*1	Internal bandwidth	Ethernet & CPU total bandwidth	Throughput (Ether2Ether)
64 KB	250 MHz	32 Gbps	MAX 1G bps (Exp 100M Ether 2 port + 100M CPU 1 port) × TX/RX	800 to 2000 ns
64 KB	50 MHz	6.4 Gbps	MAX 1G bps (Exp 100M Ether 2 port + 100M CPU 1 port) × TX/RX	2000 to 6000 ns
64 KB	250 MHz	32 Gbps	MAX 3G bps (Exp 1G Ether 2 port + 1G CPU 1 port) × TX/RX	400 to 600 ns
64 KB	150 MHz	19.2 Gbps	MAX 3G bps (Exp 1G Ether 2 port + 1G CPU 1 port) × TX/RX	500 to 800 ns

Note: At the frame end (forwarding source port) - frame start (forwarding destination port). When measuring, there is no dependency on the payload length of the frame.

Note: The frame start (forwarding destination port) is up to the point where the preamble starts to output.

Note 1. Specifications less than these are prohibited.
Throughput measurement conditions:

- From frame end (forwarding source port) to frame start (forwarding destination port)
- The forwarding (destination port) is not busy
- L2 forwarding unicast (1:1 forwarding).

30.5 Operation

30.5.1 Internal Clocks

Table 30.10 lists ESWM internal clocks. These clocks are supplied by the Clock Generation Circuit (CGC). Set the clock before starting the ESWM operation.

Table 30.10 Configurable range of clocks

VDD Voltage [V]	Speed [Mbps]	ICLK [MHz]	PCLKA [MHz]	ESWCLK [MHz]	ESWPHYCLK [MHz]
VSCR_1, voltage range 1	1000	150 to 250	up to 125	150 to 250	500
	100/10	50 to 250	up to 125	50 to 250	—
VSCR_2, voltage range 2	1000	100 to 150	up to 75	150 to 225	500
	100/10	50 to 150	up to 75	50 to 225	—

Clock restrictions are as follows:

- $ICLK \times 1.5 \geq ESWCLK$
- $ESWCLK > PCLKA$
- $ICLK > PCLKA$.

30.5.2 PHY Interface

30.5.2.1 PHY Interface Select

Table 30.11 lists the configuration settings for the ESWM PHY interface. Set the MIICRn.MIISEL[1:0] bits first before setting the MPIC register in RMACn.

Table 30.11 Configuration settings for ESWM PHY interface

Type	Speed	RMACn.MPIC		MIICRn
		LSC[2:0]	PIS[2:0]	MIISEL[1:0]
MII	10 Mbps	000b	000b (MII)	00b
MII	100 Mbps	001b	000b (MII)	00b
GMII	1 Gbps	010b	010b (GMII)	00b
RGMII	10 Mbps	000b	000b (MII)	01b
RGMII	100 Mbps	001b	000b (MII)	01b
RGMII	1 Gbps	010b	010b (GMII)	01b
RMII	10 Mbps	000b	000b (MII)	10b
RMII	100 Mbps	001b	000b (MII)	10b

30.5.2.2 MII/GMII Interface

When MIICRn.MIISEL[1:0] = 00b, MII/GMII interface is selected. GTX_CLK is generated by dividing ESWPHYCLK by 4.

30.5.2.3 RGMII Interface

ESWM can be converted from an external RGMII to an internal GMII/MII. The RGMII interface is enabled when MIICRn.MIISEL[1:0] = 01b. Since the initial state of the ESWM PHY interface is MII/GMII, it becomes operational after the reset is released with the MIIRR.RGRST bit. At this time, TXC is generated by dividing ESWPHYCLK. MIICRn. Set MIISEL[1:0] to 01b before a release reset.

30.5.2.3.1 RGMII Internal Delay Mode

The RGMII interface supports internal delay mode for RGMII_n_TXC.

30.5.2.4 RMII Interface

ESWM can be converted from an external RMII to an internal MII. The RMII interface is enabled when MIICRn.MIISEL[1:0] = 10b. Since the initial state of the ESWM PHY interface is MII/GMII, it becomes operational after the reset is released with the MIIRR.RMRST bit. Set MIICRn.MIISEL[1:0] to 10b before a release reset.

30.5.3 ESWM Core Function

See the following chapters:

- [section 31, Ethernet Message Forwarding Engine \(MFWD\)](#)
- [section 31.5.4. Fabric \(MFAB\)](#)
- [section 32, Ethernet Common Agent \(COMA\)](#)
- [section 33, Ethernet Agent \(ETHA\)](#)
- [section 34, Ethernet MAC \(RMAC\)](#)
- [section 35, Ethernet CPU Agent \(GWCA\).](#)

30.5.4 gPTP and PTP Pulse Generator

See the following chapters:

- [section 36, Ethernet Generic PTP Timer \(GPTP\)](#)
- [section 36.6. Pulse Output Timer Module \(plsotim\).](#)

30.5.5 TAS Gate Status Monitor

The signal to be output to the ET_TAS_STAn pin can be selected using the TASSTMS register. [Table 30.12](#) lists the TASSTMS register select table.

Table 30.12

MSSn[4:0]	Module	Signal
00000b to 01000b	ETHA0	eha_race_tas_gate_state[0] to eha_race_tas_gate_state[8]
10000b to 11000b	ETHA1	eha_race_tas_gate_state[0] to eha_race_tas_gate_state[8]

For details of eha_race_tas_gate_state[8:0], see [section 33, Ethernet Agent \(ETHA\)](#).

30.6 Interrupt and Event

Table 30.13 Interrupt and event output (1 of 2)

Symbol	Source flag	Description	Type
ETHER_FWEI	MFWD.FWEIS	Forwarding Error Interrupt	Level
ETHER_CAEI	COMA.CAEIS	Common Error Interrupt	Level
ETHER_GWEI	GWCA0.GWEIS0 (other than GWEIS0.TDFES), GWCA0.GWEIS1, GWCA0.GWEIS4, GWCA0.GWEIS5	GWCA0 Error Interrupt	Level
ETHER_EAEI0	ETHA0.EAEIS0, ETHA0.EAEIS1, ETHA0.EAEIS2, RMAC0.MEIS	ETHA0 Error Interrupt	Level
ETHER_EAEI1	ETHA1.EAEIS0, ETHA1.EAEIS1, ETHA1.EAEIS2, RMAC1.MEIS	ETHA1 Error Interrupt	Level
ETHER_PTPSI0	GPTP.PTPIS0	gPTP Status Interrupt 0	Level
ETHER_PTPSI1	GPTP.PTPIS1	gPTP Status Interrupt 1	Level

Table 30.13 Interrupt and event output (2 of 2)

Symbol	Source flag	Description	Type
ETHER_FWSI	MFWD.FWMIS	Forwarding Status Interrupt	Level
ETHER_CAMI	COMA.CAMIS	Common Status Interrupt	Level
ETHER_EASI0	RMAC0.MMIS	ETHA0 Status Interrupt	Level
ETHER_EASI1	RMAC1.MMIS	ETHA1 Status Interrupt	Level
ETHER_GWDI7	GWCA0.GWDIDS, GWCA0.GWEIS2, GWCA0.GWEIS3 with mapped by TOP.TPEMIMC7t	GWCA Data Interrupt	Level
ETHER_GWDI6			
ETHER_GWDI5			
ETHER_GWDI4			
ETHER_GWDI3			
ETHER_GWDI2			
ETHER_GWDI1			
ETHER_GWDI0			
ETHER_TSDI1	GWCA0.GWTSDIS, GWCA0.GWEIS0.TDFES with mapped by TOP.TPMIMC6t	Time Stamp Data Interrupt	Level
ETHER_TSDI0		Time Stamp Data Interrupt	Level
ETHER_MDIO0	RMAC0.MMIS1	MDIO Interrupt 0	Level
ETHER_MDIO1	RMAC1.MMIS1	MDIO Interrupt 1	Level
ETHER_RMPI0	RMAC0.MMIS.LPIAIS, RMAC0.MMIS.LPIDIS, RMAC0.MMIS.MPDIS	RMAC0 PHY Interrupt	Level
ETHER_RMPI1	RMAC1.MMIS.LPIAIS, RMAC1.MMIS.LPIDIS, RMAC1.MMIS.MPDIS	RMAC1 PHY Interrupt	Level
GPTP_MATCH0	—	Media Clock Recovery Match 0	Pulse
GPTP_MATCH1	—	Media Clock Recovery Match 1	Pulse
GPTP_PTPOUT0	—	PTP Pulse Out 0	Pulse
GPTP_PTPOUT1	—	PTP Pulse Out 1	Pulse
GPTP_PTPOUT2	—	PTP Pulse Out 2	Pulse
GPTP_PTPOUT3	—	PTP Pulse Out 3	Pulse

Table 30.14 Event input

Symbol	Event source	Description
ELC_GPTP0	ELSR31	gPTP Media clock capture event 0
ELC_GPTP1	ELSR32	gPTP Media clock capture event 1

The gPTP media clock capture can select event source using the PTP Capture Event Select register.

30.7 Usage Notes

30.7.1 Settings for the Module-Stop Function

The MSTPC30 bit in Module Stop Control Register C (MSTPCRC) can enable or disable the ESWM module operation. The modules are initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see [section 11, Low Power Mode](#).

30.7.2 Transition to Software Standby Mode

ESWM should run Switch disable flow to disable it before transitioning to software standby.

If RGMII interface or RMII interface is selected, these interfaces need to be reset after Switch disable flow by the MIIRR register.

30.7.3 Return from Software Standby Mode

After returning from Software Standby mode, ESWM should run the switch initialization flow to initialize it.

30.7.4 Transition to ESWM Power Domain off by Power Gating

ESWM should run Switch disable flow to disable it before transitioning to the ESWM power domain off.

If RGMII interface or RMII interface is selected, these interfaces need to be reset after switch disable flow by the MIIRR register.

30.7.5 Transition to ESWM Power Domain on by Power Gating

After the ESWM power domain is on, ESWM should run the switch initialization flow to initialize it.